SYNAPSE EDGE DEFENSE

Risk Scoring Lifecycle

How actor risk accumulates, decays, and triggers actions. Dynamic risk scoring at the edge — sub-millisecond decisions, no backend required.

0–100	40	70	−10/min
RISK RANGE	CHALLENGE THRESHOLD	BLOCK THRESHOLD	DECAY RATE

RISK SCORE THRESHOLD MODEL

0–39 ALLOW	40–69 CHALLENGE	70–100 BLOCK

0 40 100

- Below threshold — request passes, actor monitored
- Interrogator engaged — progressive challenges issued
- Blocked on sight — request rejected, tarpit active

RISK SOURCES — WHAT ADDS POINTS

Challenge Failures +35

Failed Interrogator challenge
JavaScript execution failed
CAPTCHA timeout / wrong answer

Strong indicator of bot behavior

Campaign Membership +30

Linked to active campaign
Correlated via fingerprints
Shared attack patterns

Automatic when correlated

WAF Detections +25

SQLi, XSS, path traversal
Command injection
Protocol violations

Per detection, configurable

Threat Intelligence +20

Known bad IP lists
Tor exit nodes
Datacenter / VPN ranges

Configurable feeds

Rate Violations +15

Exceeding request limits
Burst patterns
Endpoint hammering

Scales with severity

Signal Anomalies +10

TLS / User-Agent mismatch
Session behavior anomalies
Auth token inconsistencies

Per anomaly type

EXAMPLE — CREDENTIAL STUFFING ATTACK

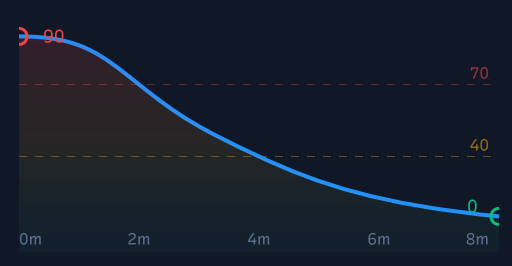
Risk climbing from 0 → 90 in 6 requests

ACTOR: 198.51.100.12 · JA4: a]b1c2d3e4f5 · SESSION: new

EVENT	DELTA	SCORE
Request 1: Rate limit triggered	+15	15
Request 5: Failed login pattern detected	+10	25
Request 10: Campaign correlation link	+30	55
CHALLENGE — Interrogator challenge sent	—	55
Challenge failed — JS PoW not computed	+35	90
BLOCKED — Request rejected · Tarpit engaged	—	0

Decay: 5 minutes of good behavior to return below CHALLENGE threshold (if behavior improves)

RISK DECAY — HOW RISK REDUCES



−10 pts/min

Configurable decay rate. Risk decreases every minute of good behavior — legitimate requests with no WAF triggers.

Persistent

Risk survives restarts. Actors can't reset their score by reconnecting or rotating IPs (fingerprint tracking).

Configurable

RISK_DECAY_RATE_PER_MINUTE adjustable via API or config file. Tune for your traffic patterns.

Reward Good Behavior

Returning users who behave see their risk drop naturally. No manual intervention needed.

Dynamic risk scoring at the edge. No backend required.

Sub-millisecond decisions. Progressive enforcement. Risk that decays with good behavior.